SUPPORTING ADULT GUIDANCE

Bearings — describing direction

Constructions, Loci and Bearings · Lesson 6. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson introduces three-figure bearings as the standard way to describe direction — clockwise from North, always three figures. Learners read and state bearings, find back bearings (±180°), and fix a position from two bearings (triangulation), which links directly to the intersection-of-loci idea from Lesson 5. A bearing line is itself a locus of a direction.

Driving Question: *How do we describe a direction precisely, so anyone means the same thing?*

The mathematics, in plain terms

A bearing is the clockwise angle from North to a direction, written with three figures (adding leading zeros). Because North lines are all parallel, the reverse direction (the back bearing) differs by exactly 180° — add 180° to a forward bearing under 180°, subtract 180° from one that is 180° or more. A single bearing only says the target lies somewhere along a line; two bearings from two known places cross at one point, fixing the position. This is how navigation and search-and-rescue locate things.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Reading a bearing

State the bearing of B from A.

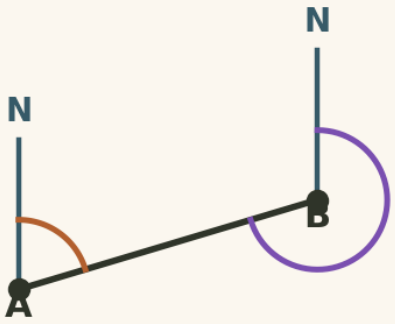


1. Draw a North line at A.	North at A
2. Turn clockwise from North to the line AB.	clockwise turn
3. Measure the angle and write three figures.	= 072°

The bearing of B from A is 072° — clockwise from North, written with three figures.

Example 2 — Finding a back bearing

The bearing of B from A is 072°. Find the bearing of A from B.



1. The North lines at A and B are parallel.	parallel Norths
2. The return faces the opposite way — a half turn.	+ 180°
3. 072° is under 180°, so add 180°.	72 + 180

The back bearing (A from B) is 072° + 180° = 252°.

Convention first, then use

The whole gain is the shared convention: clockwise, from North, three figures. Insist on all three parts every time, and on the leading zeros (072°, not 72°) — this is exactly what exams and real navigation require. Let learners read plenty of bearings on the dial before writing them, and meet back bearings by seeing the parallel North lines rather than memorising '±180'.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Meet the convention — the eight compass points and their three-figure bearings.	~9 min
Movement	Read and set bearings on a dial; hit a target bearing by dragging.	~13 min
Reflection	Discover that a back bearing differs by 180°, and see why (parallel North lines).	~9 min
Creativity	Fix a ship's position from two coastguard bearings — triangulation.	~9 min

Rest	A short pause on how a bearing can guide someone safely home.	~2 min
Nutrition	Name what the learning feeds: shared convention, and real consequence.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Writing a bearing with fewer than three figures.
072°, not 72°. Ask the learner to add leading zeros so every bearing has three figures.

Measuring anticlockwise, or from a direction other than North.
Bearings are always clockwise from North. Ask: ‘Where do we start, and which way do we turn?’

Thinking the back bearing is the same as the forward bearing.
The return faces the opposite way — a half turn, 180°. Ask the learner to compare the two North lines and the two directions.

Adding 180° when they should subtract (or vice versa).
Add 180° if the bearing is under 180°; subtract if it is 180° or more, so the answer stays between 000° and 360°.

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: the four compass directions and measuring an angle in degrees. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner writes ‘72°’ without the leading zero, or gives a back bearing that is out of range. The aim is a question that returns the reasoning.

- “How many figures should every bearing have? Show me 72° written properly.”
- “Where do we start measuring, and which way do we turn?”
- “The two North lines point the same way — so how far apart are the there-and-back directions?”

- “Is your back bearing between 000° and 360°? If not, add or subtract 360°.”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Why three figures?
So there is never any ambiguity — 072° can't be misread. It's the standard in navigation and in exams.

Do learners need a protractor?
For hand work, yes — a protractor measures the clockwise angle from North. The on-screen dial does the same job for exploring.

What's a back bearing for?
It's the direction of the return journey. If a boat sails from A to B on one bearing, the bearing to sail back (B to A) is the back bearing.

How do two bearings fix a position?
Each bearing is a line of possible positions; two lines from two known places cross at exactly one point — the fix. This is triangulation.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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